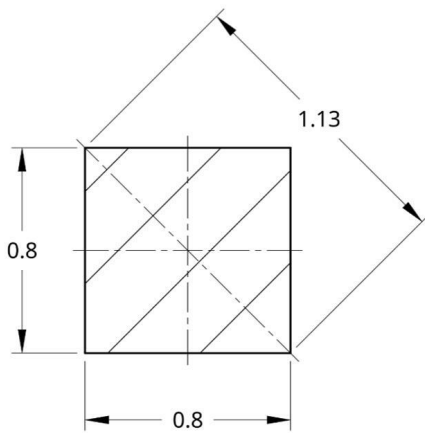


BROKEN QUALITY PRESS PRESENTS

Using GD&T to improve an automated manufacturing process

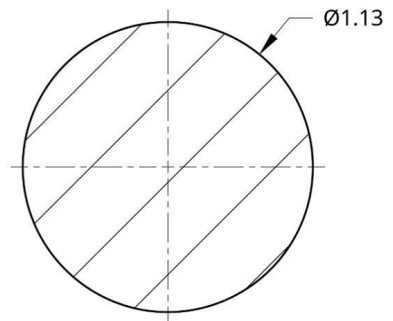
How to increase yield using positional tolerance zones

An essay by Jim Lehner



LINEAR TOLERANCE ZONE

POSITIONAL TOLERANCE ZONE



JIM LEHNER

Using GD&T to improve an automated manufacturing line

How to increase yield using positional tolerance zones

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First edition

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Chapter 1

Automated pick and place

An automated manufacturing line for hydrogen fuel cell stacks picks and places an Open Flow Field (OFF) on a dimple plate. Coordinate tolerances and symmetrical limits define acceptable placement for the corner of the OFF, with respect to the outer edge of the dimple plate's rubber gasket, to be 5.0 ± 0.4 mm in the X direction and 2.5 ± 0.4 mm in the Y direction (see Figure 1).

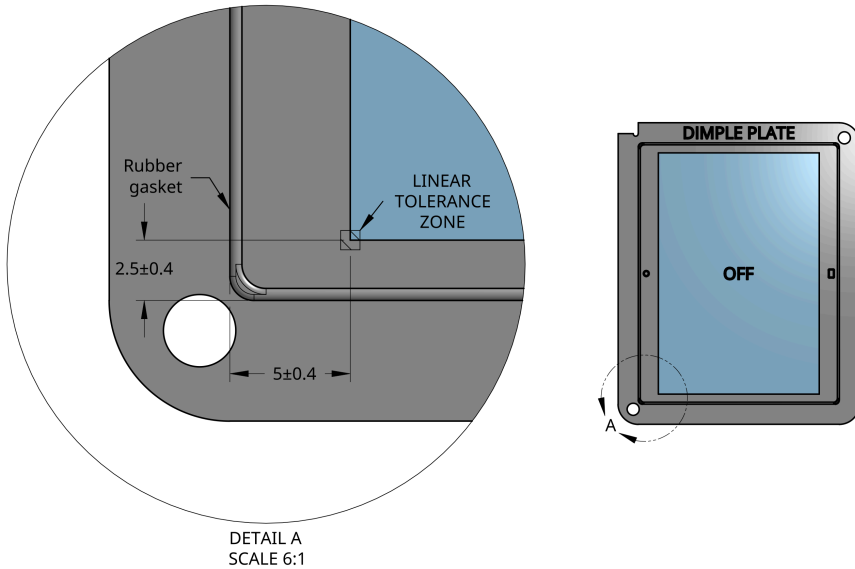


Figure 1. Assembly drawing using coordinate tolerances and symmetrical limits to control position of the OFF corner. Hatched region indicates linear tolerance zone.

The coordinate tolerances using symmetrical limits shown in Figure 1 were used as guides when implementing the logic for the line's vision system. After identifying the corner of the OFF, the system uses the outer edge of the dimple plate's rubber gasket to register the location of fiducials. With the fiducials registered and OFF corner identified, the system measures the distance between the fiducials marking the gasket edges and the OFF corner in both X and Y. Any measurement that falls within the linear tolerance zone, stretching from the coordinates (2.1, 4.6)—the lower-left corner—to (2.9, 5.4)—the upper-right corner—is accepted, while any measurement that falls outside this zone is rejected. The linear tolerance zone has a 0.8 mm width and height and an area of 0.64 mm^2 (see Figure 2).

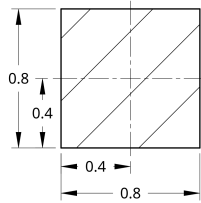


Figure 2. Linear tolerance zone with a width and height of 0.8 mm and area of 0.64 mm².

The linear tolerance zone from Figure 2 is overlaid on a scatter plot of part placement measurements in Figure 3¹. Review of the figure reveals that the process is placing parts high and to the left of the linear tolerance zone. Of the 95 measurements shown in Figure 3, 61 out of 95 (64.2%) fall within the linear tolerance zone with the remaining 34 falling outside of it. To address this placement bias, adjustments to the placement mechanism will be necessary. However, for these adjustments to be useful, the underlying causal system must be in a state of statistical control (predictable). That is, in order for adjustments to the placement mechanism to yield expected results, the underlying causal system must be influenced by only common causes of routine variation.

¹ For the Python code used to associated with this essay, visit: github.com/jimlehner/broken-quality-initiative/tree/main/code

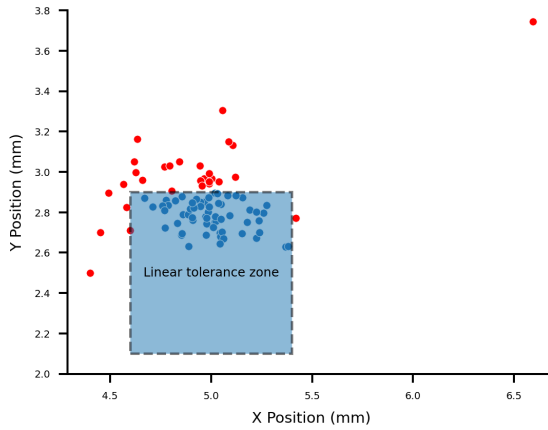


Figure 3. Scatter plot of OFF placement measurements with linear tolerance zone overlaid. Measurements outside the linear tolerance zone are red, while measurements inside are blue.

Although the scatter plot in Figure 3 provides a sense of the line's ability to place parts within the linear tolerance zone, it reveals nothing about the types of variation influencing process behavior. The only way to determine this is by using process behavior charts, specifically XmR charts.

After disaggregating part placement into its X and Y components, XmR charts for each component should be constructed and the underlying causal system characterized. If assignable causes of exceptional variation are present in either component, they must be eliminated before adjusting the placement mechanism. While few in number, assignable causes are dominant in their effect. This dominance will cause a process to bounce and shift in unpredictable ways, making adjustments more akin to guessing than a form of data-driven decision-making.

While most teams will rightfully jump into action given the findings of the scatter plot, a more foundational opportunity for improvement lies in wait. This involves updating the assembly drawing using Geometric Dimensioning and Tolerancing (GD&T). By replacing the coordinate tolerances and symmetrical limits in Figure 1 with basic dimensions and a positional

tolerance, the linear tolerance zone is replaced with a positional tolerance zone with a larger area.

Linear to positional

Coordinate tolerances and symmetrical limits yield linear tolerance zones with areas that are calculated by multiplying the tolerance zones length by its width. Basic dimensions and positional tolerances (GD&T) yield positional tolerance zones with areas that are calculated using the formula for the area of a circle: πr^2 . To transform a linear tolerance zone into a positional tolerance zone requires a little geometry.

Figure 4 shows the linear tolerance zone resulting from coordinate tolerances and symmetrical limits of 5.0 ± 0.4 mm and 2.5 ± 0.4 mm, as well as the positional tolerance zone resulting from the use of basic dimensions and the feature control frame. Note that the diagonal of the linear tolerance zone in Figure 4 has a length that is equal to the diameter of the positional tolerance zone. The relationship between these two dimensions is described by the Pythagorean Theorem.

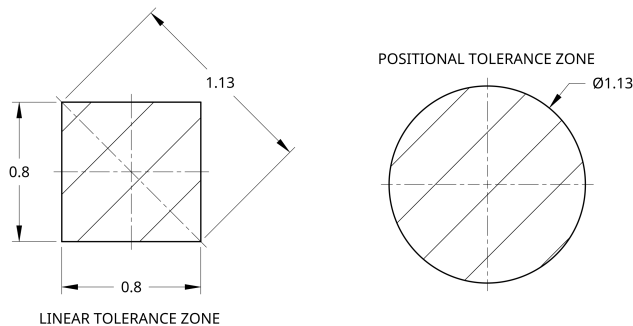


Figure 4. Linear tolerance zone from coordinate tolerances and symmetrical limits (Left) and positional tolerance zone from basic dimensions and the feature control frame (Right).

The Pythagorean Theorem relates the sides of a right triangle using the equation:

$$a^2 + b^2 = c^2$$

Here, a^2 and b^2 are the legs (the sides of the right triangle that form a right angle), while c^2 is the hypotenuse (the longest side of the right triangle). Since the length of the hypotenuse is the dimension of interest, solving the Pythagorean Theorem for c yields the formula:

$$c = \sqrt{a^2 + b^2}$$

Thus, the length of the hypotenuse (c) is equal to the square root of the sum of the squares of legs (a and b). Both sides of the linear tolerance zone have lengths of 0.8 mm. Substituting this value for a and b in the formula for c yields a diagonal length for the linear tolerance zone of 1.13 mm. This value will represent the diameter of the circle for the positional tolerance zone that is created using basic dimension and the feature control frame.

$$\begin{aligned} c &= \sqrt{0.8^2 + 0.8^2} \\ &= \sqrt{0.64 + 0.64} \\ &= \sqrt{1.28} \\ &= 1.13 \end{aligned}$$

Basic dimensions and the feature control frame

To take advantage of the increased area of a positional tolerance zone, basic dimensions will be used to define its location. Basic dimensions are shown on a drawing by enclosing the values in a rectangle. They differ from coordinate

tolerances in that they define a theoretically exact value for the size, shape, orientation, or location for a feature size.

The basic dimensions for the updated assembly drawing will reference fixed features on the dimple plate called datums. For the X axis, the datum is the outer vertical edge of the rubber gasket, with a corresponding basic dimension of 5 mm. For the Y axis, the datum is the bottom horizontal edge of the rubber gasket, with a corresponding basic dimension of 2.5 mm. These basic dimensions represent the nominal values for the coordinate tolerances shown in Figure 1.

Basic dimensions alone do not create a positional tolerance zone. It also requires the addition of the feature control frame. The feature control frame is a rectangular box on a drawing that specifies the geometric tolerance for a feature of size. It defines how much a feature can deviate from its ideal state as defined by basic dimensions.

The feature control frame is composed of five elements: geometric characteristic, tolerance zone shape, geometric tolerance value, tolerance modifier, and the datum reference frame. The feature control frame that will be used to update the assembly drawing, labeled in accordance with the five parts, is shown in Figure 6.

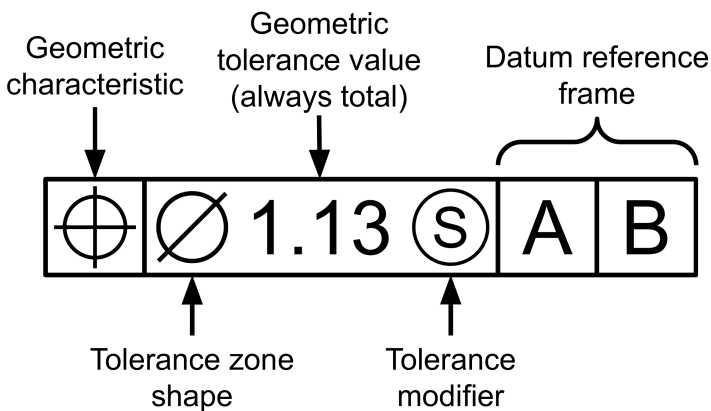
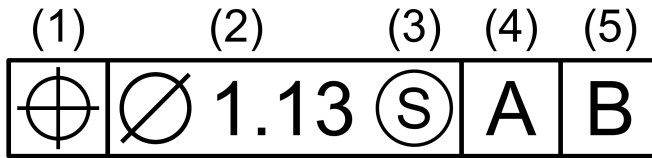


Figure 5. The feature control frame and its parts.

Assuming a reader has the requisite knowledge of its parts, interpreting the feature control frame is no different than reading a sentence. As shown in Figure 6, the feature control frame states that *the position of the OFF corner must be on or within a cylindrical tolerance zone of 1.13 mm, regardless of feature size. Datums A and B are the measuring surfaces for the 5 and 2.5 basic dimensions.*



- (1) The position of the OFF corner
- (2) Must be on or within a cylindrical tolerance zone of 1.13 mm
- (3) Regardless of feature size.
- (4) Datum A and
- (5) Datum B are the measuring surfaces for the 5 and 2.5 basic dimensions.

Figure 6. Interpretation of the feature control frame.

Updates to the drawing

The updated assembly drawing with basic dimensions and the feature control frame is shown in Figure 7. Note that the S tolerance modifier indicates that the specified geometric tolerance applies regardless of feature size (RFS). This means that the tolerance value remains constant. RFS is used because the positional tolerance governs a fixed positional tolerance zone that defines the acceptable placement of the corner of the OFF with respect to the basic dimensions.

These GD&T changes to the drawing will guide updates to the line's vision system. The increased area of the positional tolerance zone will allow for a larger number of assemblies to be accepted.

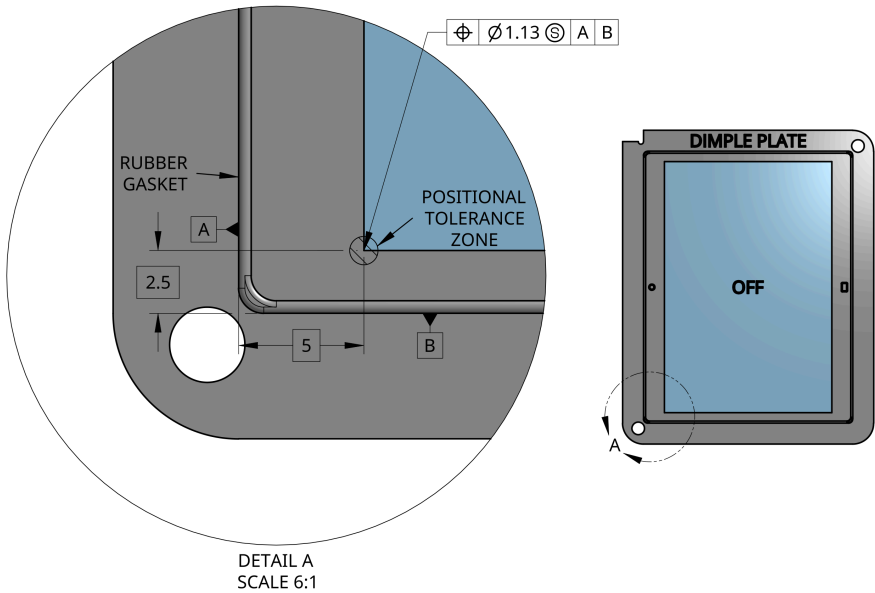


Figure 7. Assembly drawing using basic dimensions and the feature control frame to control position of the OFF corner. Hatched region indicates positional tolerance zone.

The advantages of the positional tolerance zone

Updating the assembly drawing with GD&T yields a circular positional tolerance zone that has an area of 0.985 mm^2 . This is in contrast to the square linear tolerance zone with an area of 0.64 mm^2 . The area of the positional tolerance zone in the drawing using GD&T is 53.9% larger than the area of the linear tolerance zone.

Overlaying the scatter plot of part placement measurements from Figure 3 with both the linear tolerance zone and positional tolerance zone in Figure 9, the increase in tolerance zone area becomes obvious. Where 61 out of 95 (64.2%) of the measurements fall within the linear tolerance zone, 76 out of 95 (80%) of the same measurements fall within the positional tolerance zone. Before any adjustments to the placement mechanism or analysis of the

underlying causal system have been made, 15 additional measurements fall within the positional tolerance zone that were excluded by the linear tolerance zone.

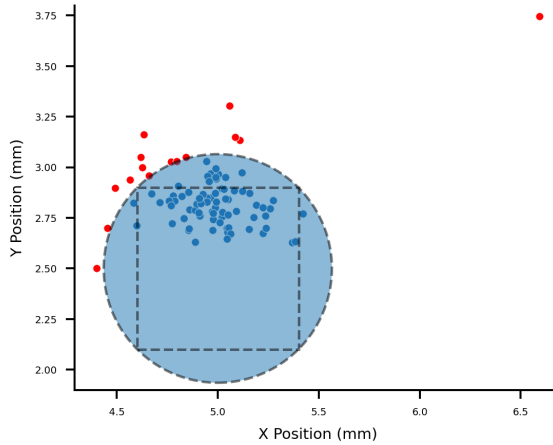


Figure 9. Scatter plot of OFF placement measurements with linear tolerance zone and positional tolerance zone overlaid. Measurements outside both tolerance zones are red, while measurements inside either zone are blue.

Summary

Questioning engineering drawings is an area where many are hesitant to tread. After all, having been checked and verified before release, it is assumed that drawings will have no errors. While the lack of GD&T for the assembly drawing in Figure 1 was certainly not an error, it did not do the automated manufacturing line any favors. By using coordinate tolerances and symmetrical limits to define the logic of the vision system, the symmetrical tolerance zone was 35% smaller than it needed to be. This resulted in scrapped assemblies that could have otherwise been accepted. While additional efforts must be made to further improve this process, optimizing the area of the

CHAPTER 1

tolerance zone using GD&T establishes a strong foundation upon which future improvements will be built. It aligns design intent with the reality that variation is an unavoidable feature of manufacturing.

